

Agentic Multimodal RAG: Low-Level Design

Module-level flow · function responsibilities · PostgreSQL schema · query path. Companion to 05-low-level-design.md (section numbers refer to it).

 Python module (laptop CPU)

 GPU job (Kaggle / Colab)

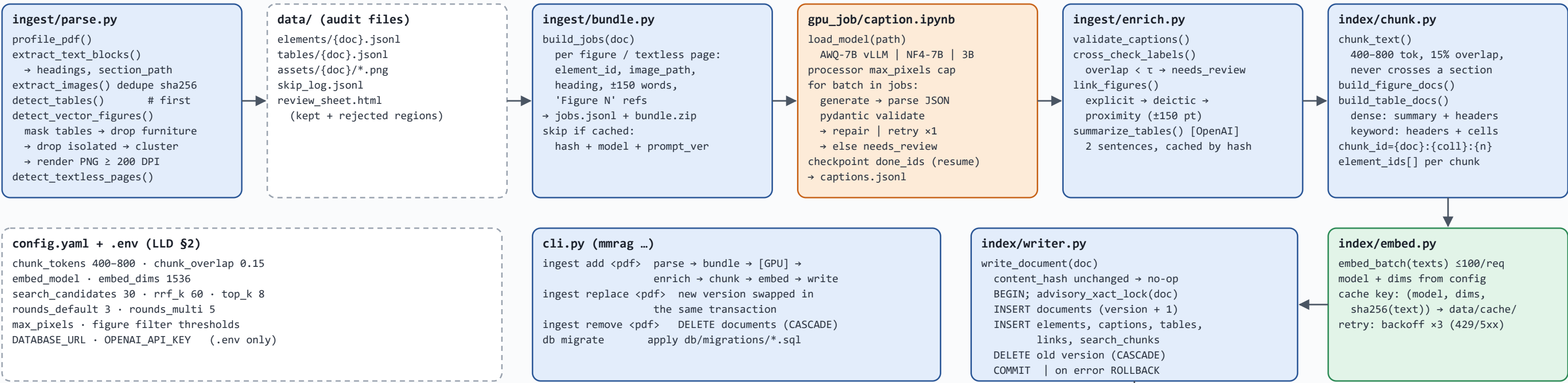
 Calls OpenAI API

 PostgreSQL table / data

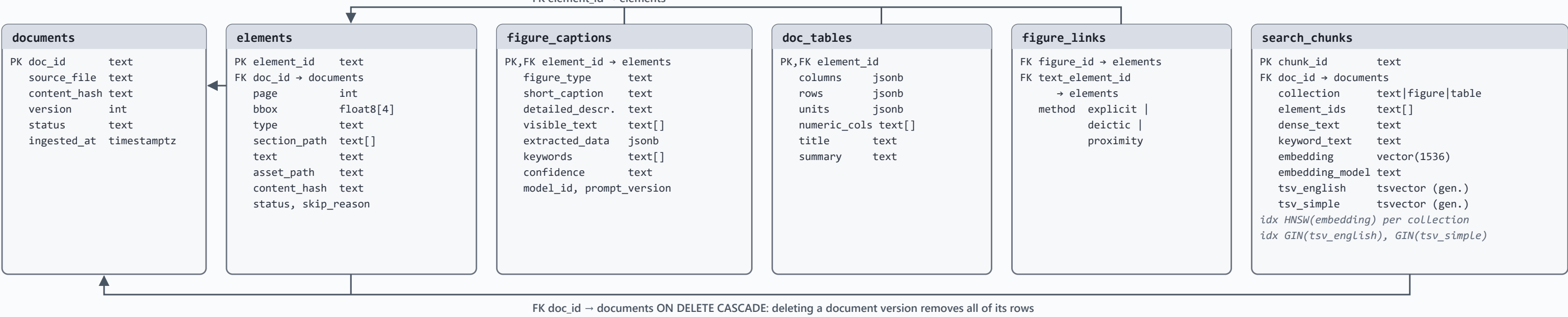
 Files on disk / config

 UI / answer

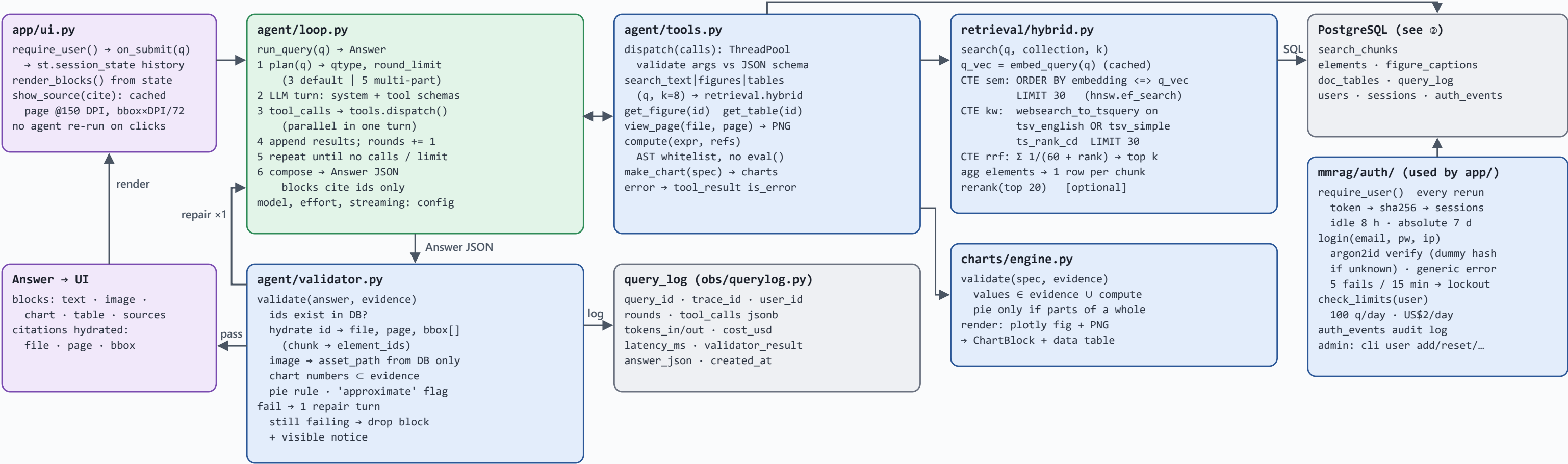
① INGESTION PATH: modules and functions (LLD §3)



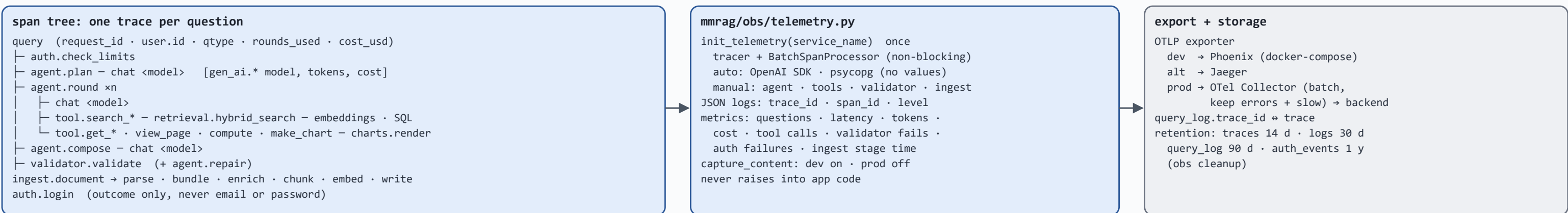
② POSTGRESQL SCHEMA: pgvector + full-text (LLD §4) · dev: pgvector/pgvector:pg16 in a named Docker volume · + users, sessions, auth_events (LLD §4.1)



③ QUERY PATH: modules and functions (LLD §5)



④ OBSERVABILITY: OpenTelemetry traces, logs, metrics (LLD §5.9)



⑤ EVALUATION: RAGAS + custom metrics (LLD §5.10)

